MASTER CLASS DELIVERABLE • VIDEO #8

20-Line Python Autonomous Agent: LangGraph, Gemini Flash & Memory

Cyclic Graph Reasoning, Dynamic Tool Calling, SQLite Checkpointer & Zero Cloud Cost

FRAMEWORK: LANGGRAPH + LANGCHAIN LLM: GEMINI 2.5 FLASH / DEEPSEEK PUBLISHED BY: @PARKER.RENDER

1. What Makes an AI Agent "Autonomous"?

Standard ChatGPT prompts are linear: you ask, it answers. An **Autonomous Agent** can plan, call external tools (execute code, search the web, query databases), observe outputs, correct mistakes in a loop, and persist memory across sessions.

2. Complete 20-Line Production Agent Script

```
from langchain_google_genai import ChatGoogleGenerativeAI
from langchain_community.tools import DuckDuckGoSearchRun
from langgraph.prebuilt import create_react_agent
from langgraph.checkpoint.sqlite import SqliteSaver
import sqlite3

# 1. Initialize Ultra-fast LLM & Tools
llm = ChatGoogleGenerativeAI(model="gemini-2.5-flash", temperature=0)
tools = [DuckDuckGoSearchRun()]

# 2. Add Persistent SQLite Session Memory
conn = sqlite3.connect("agent_memory.db", check_same_thread=False)
memory = SqliteSaver(conn)

# 3. Compile Autonomous ReAct Cyclic Graph
agent = create_react_agent(llm, tools=tools, checkpointer=memory)

# 4. Execute Multi-step Goal
config = {"configurable": {"thread_id": "session_001"}}
result = agent.invoke(
    {"messages": [("user", "Find the current stock price of NVIDIA and calculate 15% dividend yield.
    config=config
)

print(result["messages"][-1].content)
```

3. Key Production Features

- **Cyclic Decision Loop:** If a tool returns an error, the agent self-heals by rewriting the query automatically.
- **Zero Server Costs:** Runs 100% locally on your machine terminal using free-tier Gemini Flash API.
- **Threaded Checkpointing:** Resumes conversation context anytime using `thread_id`.

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